

Exp: 3b1
23/07/26

HUB

Aim:

To study the behavior of a hub observing it forward data packet using cisco packet Tracer.

Steps:

* Open Cisco Packet Tracer
* Drag and Place 1 Hub and 4 Generic PCs into the workspace
* Connect all PCs to hub using Copper Straight-Through Cables
* Select Add simple PDU from common tools bar
* Send a packet from one PC to another PC
* Switch to simulation Mode and observe how fast hub forward the packet.

Result:

The hub broadcast the received packet to all connected devices. Every Pc receives the pack only the destination pc accept it.

EXP 3b2

SWITCH

To study the behavior of a switch by observing how it forward data packet using cisco packet.



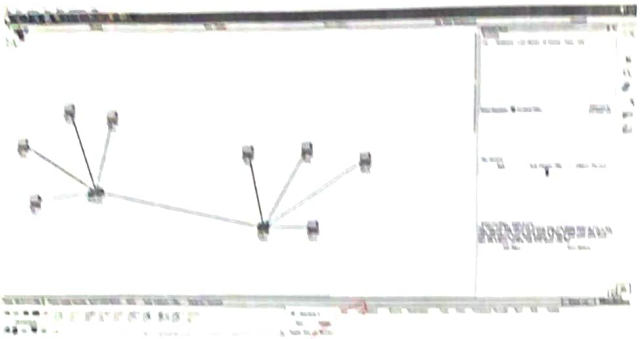




* Open cisco packet Tracer
* Drag and place 1 switch and 4 Generic PCs into workspace
* Connect all PCs to all switch using copper cable
* Select Add simple PDU from commol - tool bar
* Send a packet from one PC to another
* Switch to simulation mode and observe

Result:

The switch forward the packet only to the destination PC by using its MAC address table. This reduces unencessary network traffic.



EXP: 8&9

HUB AND SWITCH

Aim:

To compare the packet forwarding behavior of a hub and switch using cisco packet Tracer.

Steps:

* Create one network using a hub and another using switch.
* Configure IP address for all PCs in both network.
* Send a simple PDU between two pcs in each network.
* observe packet transmission in simulation mode.
* Compare how the hub and switch forward packet.

Result:

The hub broadcast packet to all connected devices, whereas the switch forwards only to packet. Therefore Switch provide better performance.